



Bangladesh Mathematics and Science Olympiad

## **BdMSO Syllabus 2026**

May 7, 2026

### **Organizer**

Society for the Popularization of Science, Bangladesh (SPSB)  
Bangladesh Open Source Network (BdOSN)

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**1. Arithmetic and Number Theory**

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| A. Whole & Rational Numbers | Absolute Value, understanding fractions, decimals and percentages, and converting between them.                                |
| B. Properties of Numbers    | Even and odd integers, prime and composite numbers.                                                                            |
| C. Factors and Multiples    | Greatest Common Divisor (GCD) and Least Common Multiple (LCM).                                                                 |
| D. Divisibility & Sequences | Divisibility rules, remainders, cyclic patterns, ordering of numbers, and evaluating complex arithmetic & geometric sequences. |
| E. Ratio and Proportion     | Solving real-world problems using ratios and direct/inverse proportions.                                                       |

**2. Geometry and Spatial Sense**

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| A. Polygons & Circles                     | Angle relationships, properties of triangles, quadrilaterals, parallelograms, trapeziums, and circles.                                                   |
| B. Measurements                           | Calculating angles, area, and perimeter of various 2D shapes.                                                                                            |
| C. Coordinate & Transformational Geometry | Basics of the Cartesian plane, plotting points, grid distances, and spatial transformations: symmetry, reflection, rotation, similarity, and proportion. |
| D. 3D Geometry                            | Properties of solid figures, cubes, parallelepipeds.                                                                                                     |
| E. Spatial Visualization                  | Understanding and solving problems involving the nets of 3D shapes.                                                                                      |

**3. Algebra and Pattern Recognition**

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| A. Pattern Recognition  | Identifying and predicting terms in numeric and geometric sequences.                          |
| B. Basic Algebra        | Evaluating algebraic expressions, variables as unknowns, and solving simple linear equations. |
| C. Advanced Expressions | Introduction to quadratic expressions and basic factorization                                 |

**4. Data Analysis, Statistics, and Measurement**

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|-------------------------|------------------------------------------------------------------------------------------|
| A. Data Representation  | Reading and interpreting tables, bar charts, pie charts, line graphs, and scatter plots. |
| B. Statistical Averages | Mean, Median, and Mode.                                                                  |
| C. Advanced Averages    | Understanding how changing individual values in a dataset affects the overall average    |

**5. Combinatorics and Probability**

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| A. Counting Techniques | Systematic counting, finding the number of possible ordered pairs, and basic permutations/combinations, and the Inclusion-Exclusion principle. |
| B. Probability         | Calculating the likelihood of specific outcomes in random events, understanding independent and dependent events.                              |

**6. Recreational Mathematics**

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| A. Mathematical Puzzles | Logic grids, parity, magic squares, and critical thinking brainteasers. |
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**1. Scientific Method and Experimental Reasoning**

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|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Scientific Thinking                       | Observation, forming hypotheses, and drawing conclusions based on evidence.                                                                                              |
| B. Measurements                              | Understanding units, length, mass, time, temperature, volume. Using scientific instruments, Accuracy, precision, and estimation.                                         |
| C. Experimental Design & Data Interpretation | Fair testing, independent, dependent, and controlled variables. Organizing data in tables, interpreting experimental diagrams. Analyzing graph-based reasoning problems. |

**2. Life Science**

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| A. The Basics of Life            | Characteristics of living vs. non-living things, The cell as the basic unit of life, movement of materials in/out.                                                                                                                                                                                                                                              |
| B. Plant Science                 | Photosynthesis, respiration, seed germination, pollination, tissue systems, and basic hormone regulation.                                                                                                                                                                                                                                                       |
| C. Human Anatomy & Organ Systems | Support & Movement, Skeleton, joints, and muscles.<br>Control & Coordination, Brain, spinal cord, nerves, reflex actions.<br>Sense Organs, Eyes, Ears, Nose. Skin. Skin & Temperature.<br>Circulation & Respiration, Blood components, heart, lung function, gas exchange.<br>Digestion & Excretion, Nutrient absorption, renal system, kidneys, waste removal. |
| D. Health & Metabolism           | General health, nutrition, release of energy from food, common diseases, prevention, and hygiene.                                                                                                                                                                                                                                                               |
| E. Microbiology & Genetics       | Beneficial vs. harmful bacteria, viruses, antibiotic resistance, microbes, basic DNA structure, inheritance, family traits, Introduction to Punnett Squares, evolution.                                                                                                                                                                                         |
| F. Ecology & Animal Behavior     | Habitats, food chains, food webs, ecosystems, population, lifecycle. Animal behavior, migration, hibernation, fixed action patterns. Classifying organisms based on food, anatomy, systematics, reproduction and its habitat.                                                                                                                                   |

**3. Physical Science**

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| A. Chemistry Fundamentals      | Atoms and molecules, mole, common elements, simple chemical formulas, carbon as the building block of life, and chemical reactions in everyday life. |
| B. Matter & Fluids             | States of matter, physical, chemical & biological transformation. Motion, speed, density. Properties of static fluids and gases.                     |
| C. Forces & Motions            | Gravitational, magnetic, elastic, and frictional forces. Changes in shape due to force.                                                              |
| D. Energy & Thermal Properties | Kinetic/potential energy, energy conversion, renewable energy. Temperature, heat transfer, conduction, convection, radiation.                        |
| E. Light & Sound               | Properties of light, color, vision. The mechanics of sound/vibration.                                                                                |
| F. Electricity & Magnetism     | Basic applications and models.                                                                                                                       |

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| <b>4. Earth and Solar System</b>              |                                  |                                                                                                                                                       |
| A.                                            | The Solar System                 | The Sun, planets, asteroids, comets. Earth's rotation/revolution, Earth-Moon system, phases of the moon, eclipses. Natural and artificial satellites. |
| B.                                            | Earth System                     | Structure and formation of the Earth, air, weather, climate, wind, seasons, gravitation, water cycle, and carbon cycle.                               |
| C.                                            | Natural Calamities               | Understanding extreme natural events including Earthquakes, Tsunamis, Cyclone and weather patterns like El Niño and La Niña.                          |
|                                               |                                  |                                                                                                                                                       |
| <b>5. Geography and Environmental Science</b> |                                  |                                                                                                                                                       |
| A.                                            | Global Geography & Biomes        | Continents, oceans, major biomes, deserts, rainforests, tundras. understanding basic maps, latitude, and longitude.                                   |
| B.                                            | Geography & Bangladesh           | Understanding the Bengal Delta and how it was formed, the major river networks, the Haor basin, the Sundarbans, and the significance of monsoons.     |
| C.                                            | Environmental Chemistry & Issues | Greenhouse gases, the science of the ozone layer, acid rain, methods of water purification and pollution management.                                  |
| D.                                            | Conservation & Climate Action    | Protecting local ecosystems, human-wildlife relations and conflict. Deforestation, Climate Change, saving Bangladesh's rivers, wetlands and forests.  |
|                                               |                                  |                                                                                                                                                       |
| <b>6. Biotechnology</b>                       |                                  |                                                                                                                                                       |
| A.                                            | Biotechnology Basics             | Simple fermentation, basics of genetic engineering/cloning, GMOs, and Biofuel.                                                                        |
| B.                                            | Modern Application               | Understanding how emerging scientific fields impact the world and environment                                                                         |